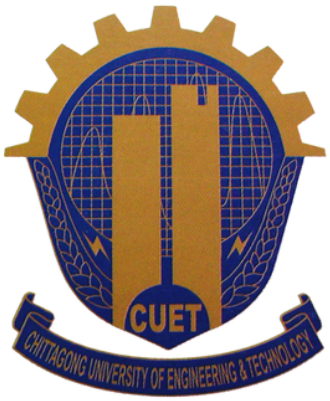
Bachelor of Science in Computer Science & Engineering



CUET ClubHub

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1 Introduction

1.1 Overview

Extracurricular activities and student organizations play an important role in supporting the academic curriculum, giving students opportunities to develop skills and build networks. Universities have many clubs that organize different events throughout the year. However, event information is often scattered across multiple social media pages and channels, making it hard to stay updated.

This fragmentation leads to several problems: students may miss events that interest while the organizers find it difficult to reach the right audience, which reduces participation.

CUET ClubHub solves these problems with a centralized platform that uses machine learning and rule-based approaches for personalized recommendations. By looking at students' interests and past participation, the system connects organizers with the most relevant participants.

1.2 Problem Statement

Educational institutions face several interconnected challenges in managing campus activities effectively:

- **Discovery Gap:** Events relevant to individual student interests and career goals frequently go unnoticed due to the absence of personalized filtering and recommendation mechanisms.
- **Administrative Burden:** Club managers and event organizers lack centralized tools for event creation, member management, registration tracking, and attendance monitoring.
- **Data Fragmentation:** Event participation data and student engagement metrics remain scattered across disconnected systems which prevents meaningful analysis.

1.3 Objectives

The primary objectives of this project is to implement a campus event management platform with the following specific goals:

- To develop a cross-platform mobile application using Flutter framework, ensuring native-like performance and user experience on both Android and iOS operating systems.

- To create accessible user interfaces optimized for distinct user roles: students seeking events and club administrators managing events.
- To achieve meaningful recommendation quality with relevance scores consistently.

2 System Architecture and Design

2.1 System Overview

The CUET ClubHub system follows a modern three-tier architecture pattern optimized for scalability, maintainability, and separation of concerns:

1. **Presentation Layer:** A Flutter-based cross-platform mobile application that provides native-like user experiences on both Android and iOS platforms. This layer handles all user interactions, local state management, and renders dynamic user interfaces based on real-time data.
2. **Application Layer:** Firebase cloud services provide authentication, real-time database operations, and cloud functions. A dedicated Flask-based RESTful API serves as the recommendation engine, processing machine learning computations independently from the main Firebase infrastructure.
3. **Data Layer:** Cloud Firestore NoSQL database stores all persistent data including user profiles, club information, event details, and registration records.

2.2 Technology Stack

- **Frontend:** Flutter
- **Backend:** Firebase
- **Database:** Firestore (NoSQL)
- **Authentication:** Firebase Authentication
- **Recommendation Engine:** Python (scikit-learn, pandas, numpy)

2.3 System Design Diagrams

2.3.1 Entity-Relationship Diagram

The database schema consists of six primary entities with carefully designed relationships to support all system functionalities while maintaining data integrity:

Primary Entities:

1. **USER** - Base authentication entity containing credentials and role information
2. **STUDENT** - Extended student profile with academic details and preferences
3. **CLUB** - Club organization information and metadata
4. **EVENT** - Event details including scheduling and location
5. **EVENT_REGISTRATION** - Junction table linking students to registered events
6. **JOIN_REQUEST** - Junction table for student-club membership requests

Key Relationships:

- USER ↔ STUDENT (1:1 mandatory relationship for student accounts)
- USER ↔ CLUB (1:1 mandatory relationship for club manager accounts)
- CLUB ↔ EVENT (1:N, one club organizes multiple events)
- STUDENT ↔ EVENT_REGISTRATION ↔ EVENT (M:N many-to-many)
- STUDENT ↔ JOIN_REQUEST ↔ CLUB (M:N with status tracking)
- STUDENT ↔ CLUB (M:N via denormalized joinedClubs array for performance)

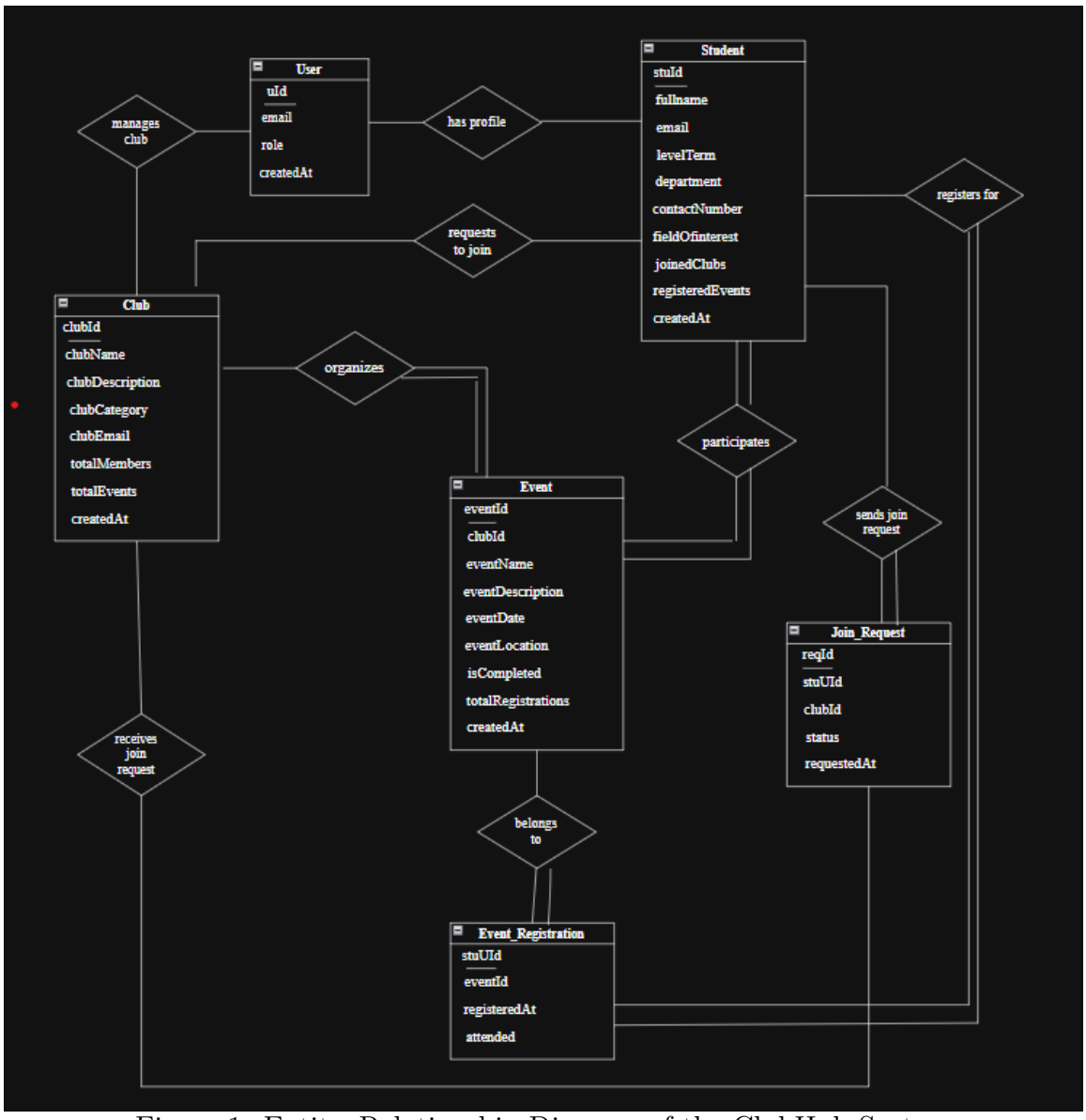


Figure 1: Entity–Relationship Diagram of the ClubHub System

2.3.2 Use Case Diagram

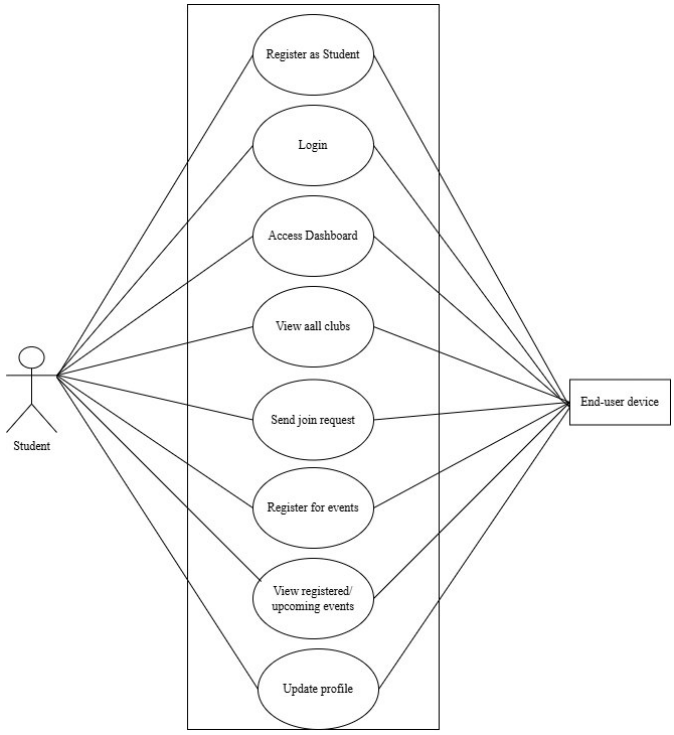
The system supports two primary actors (Student and Club Manager) with distinct but complementary use cases:

Student Actor Use Cases:

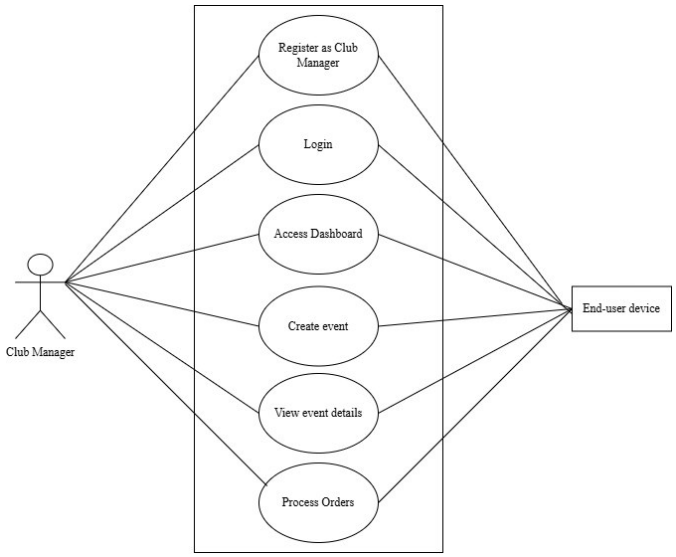
- Authentication: Sign Up, Login, Logout
- Profile Management: Update Profile, Modify Interests
- Event Discovery: Browse All Events, View Personalized Recommendations
- Event Participation: Register for Events, Unregister from Events, View Registered Events
- Club Interaction: Browse Clubs, Send Join Requests, View Joined Clubs, Leave Clubs

Club Manager Actor Use Cases:

- Authentication: Sign Up, Login, Logout
- Event Management: Create Events, Edit Events, Delete Events, Mark Events as Completed
- Registration Tracking: View Event Registrations, Export Attendance Lists
- Membership Management: Manage Join Requests ,Manage Club Members



(a) Student Usecase



(b) Club Usecase

Figure 2: Student and Club Usecase

2.4 Database Schema

2.4.1 Collections Structure

The Firestore database implements six primary collections with denormalized structures optimized for read performance:

1. Users Collection

```
{
  "uid": "unique_user_id_from_firebase_auth",
  "email": "user@example.com",
  "role": "student",  // or "club_manager"
  "createdAt": "2026-01-15T10:30:00Z"
}
```

2. Students Collection

```
{
  "uid": "unique_student_id",
  "fullName": "Student Full Name",
  "email": "u2104xxx@student.cuet.ac.bd",
  "department": "CSE",  // CSE, EEE, ME, IPE, CE, etc.
  "levelTerm": "4-1",
  "contactNumber": "+8801XXXXXXXXX",
  "fieldOfInterest": [
    "Technology & Programming",
    "Academic Research",
    "Sports & Fitness"
  ],
  "joinedClubs": ["club_id_1", "club_id_2"],
  "registeredEvents": ["event_id_1", "event_id_2", "event_id_
   3"],
  "createdAt": "2026-01-15T10:30:00Z"
}
```

3. Clubs Collection

```
{
  "clubId": "unique_club_id",
  "clubName": "Computer Club CUET",
  "clubCategory": "Technology",
  "clubDescription": "A club dedicated to promoting
   technology education and programming skills among students
```

```
    through workshops, hackathons, and tech talks.",
  "clubEmail": "computerclub@cuet.ac.bd",
  "adminUid": "club_manager_user_id",
  "totalMembers": 45,
  "totalEvents": 12,
  "createdAt": "2025-09-01T08:00:00Z"
}
```

4. Events Collection

```
{
  "eventId": "unique_event_id",
  "clubId": "club_id_reference",
  "eventName": "Inter-University Hackathon 2026",
  "eventDescription": "A 24-hour coding competition where
   teams build innovative solutions to real-world problems.
   Prizes for top 3 teams and special awards for best UI/UX
   and most innovative solution.",
  "eventDate": "2026-02-15T09:00:00Z",
  "eventLocation": "Main Auditorium, CUET",
  "eventCategory": "Technology",
  "isCompleted": false,
  "totalRegistrations": 45,
  "createdAt": "2026-01-10T14:20:00Z"
}
```

5. Event Registrations Collection

```
{
  "registrationId": "unique_registration_id",
  "studentUid": "student_user_id",
  "eventId": "event_id_reference",
  "registeredAt": "2026-01-18T16:45:00Z",
  "attended": true  // Set to true after event if student
   attended
}
```

6. Join Requests Collection

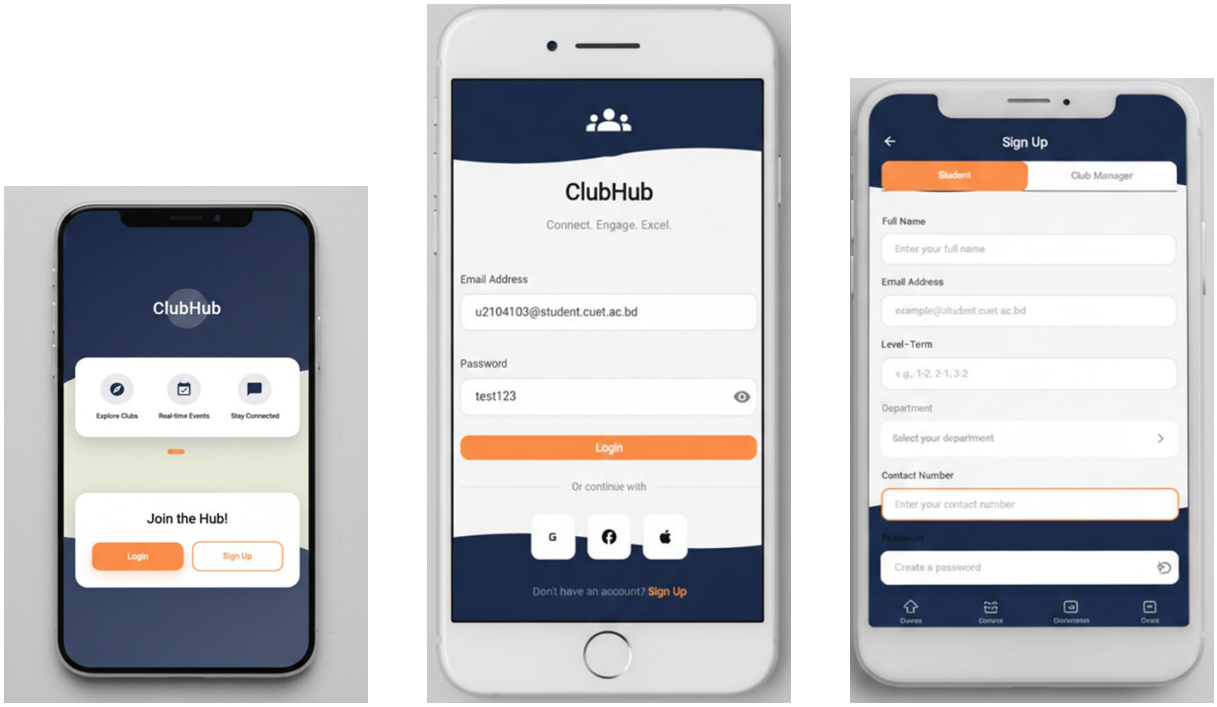
```
{
  "requestId": "{studentUid}_{clubId}_{timestamp}",
  "studentUid": "student_user_id",
```

```
  "clubId": "club_id_reference",
  "status": "pending",  // pending | accepted | rejected
  "requestedAt": "2026-01-17T11:30:00Z",
  "processedAt": null  // Set when accepted/rejected
}
```

2.5 User Interface Design

2.5.1 Authentication Screens

The application provides intuitive authentication flows with modern, accessible design following Material Design 3 principles:

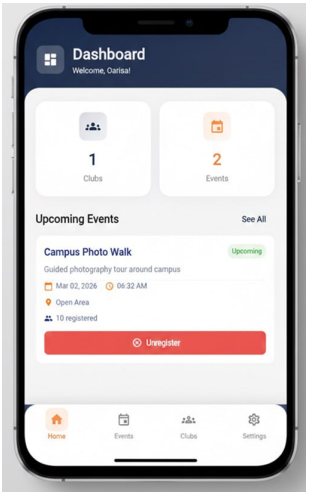


(a) Home Screen (b) Login Interface (c) Student Registration

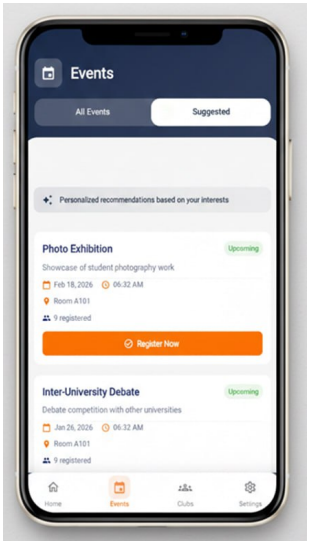
Figure 3: Authentication User Interface Screens

2.5.2 Student Interface Screens

Students access four primary screens through an intuitive bottom navigation bar, each optimized for specific tasks:



(a) Student Dashboard



(b) AI-Based Event Recommendations

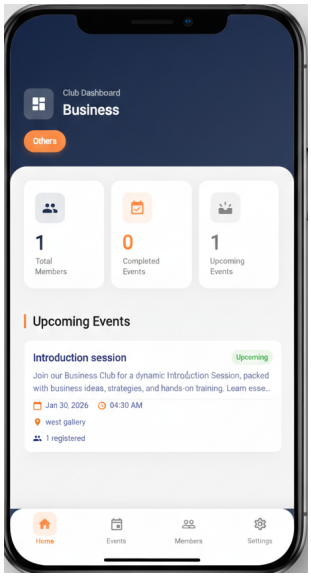
Figure 4: Student Interface Screens: Dashboard and Personalized Recommendations

Detailed Screen Descriptions:

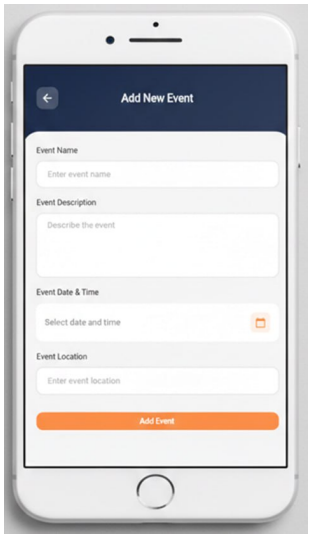
- **Dashboard:** Displays all upcoming events in chronological order with visual cards showing event name, organizing club, date/time, location, and current registration status. Students can register or unregister with a single tap.
- **Suggestions:** Shows personalized event recommendations powered by the machine learning algorithm. Each recommendation includes a relevance score and human-readable explanation tags such as "Matches your interests" or "Similar to events you attended."
- **Clubs:** Browse and join clubs with category filtering (Technology, Sports, Arts, etc.). Each club card displays member count, description preview, and action buttons for joining or viewing full details.
- **Settings:** Update personal profile information, modify interest preferences, view currently joined clubs and registered events, and access account management functions.

2.5.3 Club Manager Interface Screens

Club administrators access three primary management screens designed for efficient event and membership administration:



(a) Club Dashboard



(b) Adding new events

Detailed Screen Descriptions:

- **Dashboard:** Provides overview of all club events with key statistics (total events, total registrations). Each event card shows registration count and provides quick access to edit, delete, or mark as completed. Floating action button enables rapid event creation.
- **Join Requests:** Centralized interface for reviewing pending membership applications. Each request displays student details including name, department, and academic level to help administrators make informed decisions.
- **Create Event:** Comprehensive form for event creation with validation ensuring all required information is provided. Date and time pickers ensure proper formatting, while category selection enables better event discovery.

3 Recommendation System

The recommendation engine employs a hybrid content-based filtering approach that adapts to different user experience levels, addressing the cold-start problem. Unlike pure collaborative filtering approaches that require extensive user interaction data to function effectively, our system is architected to provide meaningful, personalized recommendations even for first-time users by heavily weighting declared interests and semantic event content similarity.

The system synthesizes three complementary techniques into a unified scoring framework:

1. **Interest-Based Matching:** Keyword-driven semantic matching between student-declared interests and event content descriptions

2. **Content Similarity:** TF-IDF vectorization with cosine similarity computation for users with established event attendance history

3. **Temporal and Quality Signals:** Time-based urgency weighting and crowd-sourced popularity metrics

3.1 Algorithm Components

3.1.1 Interest Matching (Content-Based)

The system maintains a comprehensive manually-curated keyword dictionary mapping 10 distinct interest categories to domain-specific terminology. Each category contains 20-40 carefully selected keywords representing common terms, concepts, and topics frequently appearing in related event descriptions.

```
interest_keywords = {
    'Technology & Programming': [
        'programming', 'code', 'coding', 'hackathon',
        'software', 'algorithm', 'ai', 'ml',
        'machine learning', 'web', 'development',
        'python', 'java', 'javascript', 'database',
        'cybersecurity', 'network', 'iot', 'blockchain',
        'data science', 'app development'
    ],
    'Academic Research': [
        'research', 'academic', 'paper', 'study',
        'conference', 'workshop', 'seminar',
        'presentation', 'methodology', 'analysis',
        'publication', 'thesis', 'journal'
    ],
    # ... 8 additional categories
}
```

Listing 1: Interest Keyword Mapping Example

Scoring Formula: For each event-student pair, the system counts keyword matches from the student's declared interests appearing in the event's searchable content (name, description, category):

$$\text{InterestScore} = \min\left(1.0, \frac{\sum \text{KeywordMatches}}{\sum \text{PossibleKeywords}} \times 6\right) \tag{1}$$

The multiplier (6x) provides aggressive signal amplification, particularly critical for cold-start users where interest matching represents the primary signal. The minimum function ensures scores remain bounded within [0, 1].

3.1.2 TF-IDF Vectorization

Event descriptions undergo preprocessing and transformation into numerical vector representations using Term Frequency-Inverse Document Frequency (TF-IDF), enabling mathematical computation of semantic content similarity.

Preprocessing Pipeline:

1. Combine event name (3x weight), description (2x weight), club name (1x), and category (1x)
2. Convert all text to lowercase for case-insensitive matching
3. Tokenize text into individual terms
4. Remove common English stop words (the, and, is, etc.)
5. Generate both unigrams (single words) and bigrams (two-word phrases)
6. Apply TF-IDF transformation to create weighted vector representations

```
vectorizer = TfidfVectorizer(
    max_features=500,       # Limit vocabulary size for efficiency
    stop_words='english',   # Remove common words
    ngram_range=(1, 2),     # Include unigrams and bigrams
    min_df=1,               # Minimum document frequency
    max_df=0.85,            # Maximum document frequency (85%)
    sublinear_tf=True       # Apply log scaling to term frequency
)
event_vectors = vectorizer.fit_transform(
    events_df['searchable_content']
)
```

Listing 2: TF-IDF Configuration

3.1.3 Content Similarity Calculation

For returning users with established event attendance history, the system computes similarity between candidate events and previously attended events using cosine similarity metrics:

$$\text{CosineSim}(e_i, e_j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\|\|\mathbf{v}_j\|} \tag{2}$$

Where $\mathbf{v}_i$ and $\mathbf{v}_j$ represent TF-IDF vector representations for events e_i and e_j. For each candidate event, we compute similarity to all previously attended events and aggregate the top 3 similarity scores:

$$\text{ContentSim}(e, A) = \frac{1}{3} \sum_{i=1}^{3} \text{Top}_i(\{\text{CosineSim}(e, a) \mid a \in A\}) \tag{3}$$

Where A represents the set of attended events for the given student.

3.1.4 Temporal Scoring

Events receive temporal weighting based on proximity to current date, encouraging timely discovery and reducing prominence of distant future events:

$$\text{TemporalScore}(d) = \begin{cases} 1.0 & \text{if } d \leq 7 \text{ days} \\ 0.9 & \text{if } 7 < d \leq 14 \text{ days} \\ 0.7 & \text{if } 14 < d \leq 30 \text{ days} \\ 0.5 & \text{if } 30 < d \leq 60 \text{ days} \\ 0.3 & \text{if } d > 60 \text{ days} \end{cases} \tag{4}$$

Where d represents the number of days until the event's scheduled date.

3.1.5 Quality Metrics

Event quality estimation derives from crowd-sourced popularity signals based on registration counts:

$$\text{PopularityScore}(e) = \frac{\text{Registrations}(e)}{\max_i \text{Registrations}(e_i)} \tag{5}$$

$$\text{QualityScore}(e) = \begin{cases} 0.6 & \text{if no registration data available} \\ \text{PopularityScore}(e) & \text{otherwise} \end{cases} \tag{6}$$

Newly created events without registration history receive a neutral score of 0.6 to prevent unfair penalization compared to established events.

3.2 Adaptive Score Weighting

The final recommendation score employs adaptive weighting strategies that automatically adjust based on user experience level and available behavioral data:

Cold-Start Users (0 attended events):

$$\text{Score}_{\text{new}} = 0.85 \cdot w_i + 0.08 \cdot w_t + 0.07 \cdot w_q \tag{7}$$

Returning Users (1+ attended events):

$$\text{Score}_{\text{experienced}} = 0.55 \cdot w_i + 0.30 \cdot w_c + 0.08 \cdot w_t + 0.07 \cdot w_q \tag{8}$$

Where: w_i = interest matching score, w_c = content similarity score, w_t = temporal relevance score, w_q = quality metric score.

Design Rationale:

- Cold-start users depend almost entirely (85%) on declared interests since no behavioral data exists to inform content-based or collaborative filtering.
- Experienced users benefit from balanced combination of interests (55%) and learned content similarity (30%) derived from attendance patterns, enabling personalization refinement over time.
- Temporal and quality signals remain consistent across both user segments, providing stable secondary signals.

3.3 Diversity Enforcement

To prevent recommendation homogeneity and filter bubble effects, the system applies progressive diversity penalties:

```
# Track club and category representation
club_counts = Counter()
category_counts = Counter()

for event in sorted_recommendations:
    club = event['clubId']
    category = event['eventCategory']

    # Calculate diversity penalty
    diversity_penalty = (
        club_counts[club] * 0.2 +
        category_counts[category] * 0.15
    )

    # Apply penalty (max 50% reduction)
    adjusted_score = score * (1 - min(penalty, 0.5))

    # Skip if over-represented (more than 2 from same club)
    if club_counts[club] >= 2:
        continue

    club_counts[club] += 1
```

```
    category_counts[category] += 1
```

Listing 3: Diversity Algorithm

Events from over-represented clubs or categories receive progressive score reductions up to 50%, ensuring recommendations span multiple organizational sources and thematic categories.

3.4 API Design

The recommendation service exposes a RESTful API endpoint for integration with the mobile application:

Endpoint: `GET /api/recommendations/<student_uid>`

Query Parameters:

- `top_n`: Number of recommendations to return (default: 5, maximum: 20)

Response Format:

```
{
  "studentUid": "student_123",
  "studentName": "Student Full Name",
  "interests": ["Technology & Programming", "Academic
   Research"],
  "recommendationType": "hybrid",
  "attendedEventsCount": 3,
  "timestamp": "2026-01-21T10:30:00Z",
  "recommendations": [
    {
      "eventId": "event_456",
      "eventName": "Inter-University Hackathon 2026",
      "clubName": "Computer Club CUET",
      "eventDate": "2026-02-15T09:00:00Z",
      "eventLocation": "Main Auditorium",
      "score": 0.7810,
      "componentScores": {
        "interest": 0.85,
        "content": 0.42,
        "temporal": 0.90,
        "quality": 0.65
      },
      "reasons": [
        "Matches your interest in Technology & Programming",
```

```
        "Similar to events you attended",
        "Happening soon"
      ]
    }
  ]
}
```

4 Implementation Details

4.1 Student User Flow

The complete student interaction workflow proceeds as follows:

1. User authenticates using university email credentials through Firebase Authentication
2. System retrieves student profile including declared interests and attendance history from Firestore
3. Upon accessing home screen, Flutter application makes asynchronous API call to recommendation service
4. Recommendation engine processes request:
   - Fetches student's attended events based on registration history
   - Retrieves all upcoming events from Firestore database
   - Filters out already-registered events from candidate pool
   - Calculates multi-component scores for each candidate event
   - Applies diversity algorithm to ensure variety
   - Returns top-N recommendations with explanation strings
5. Application renders personalized event cards with visual indicators
6. Student browses recommendations, views event details, and registers/unregisters with single-tap interaction
7. Registration updates propagate in real-time to Firestore and reflect immediately across all connected devices

4.2 Future Work

Potential enhancements for subsequent iterations:

- Deep learning models (BERT embeddings) for improved semantic understanding
- Collaborative filtering integration once sufficient user interaction data accumulates
- Push notification system for event reminders and personalized updates
- Calendar application integration (Google Calendar, Apple Calendar)

5 Conclusion

CUET ClubHub makes it easier for students to discover events that match their interests while helping organizers reach the right audience. By consolidating all event information and providing smart recommendations, the platform encourages participation, supports personal growth, and reduces the workload for organizers, creating a more connected community.